

The characteristic function of rough Heston models

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ANALYTICAL ABSTRACT

Rough volatility models, in which instantaneous volatility is driven by a fractional Brownian motion with a Hurst index near 0.1, reproduce both historical volatility series and implied volatility surfaces, but their non-Markovian structure removes the Fourier pricing tools that made the Heston model so widely used. The paper asks whether a Heston-type semi-closed formula for the characteristic function of the log-price survives when the Brownian kernel in the variance equation is replaced by a power kernel. The authors build a sequence of nearly unstable bivariate Hawkes processes whose rescaled limit is the rough Heston model, compute the characteristic function of the pre-limit processes through a branching (cluster) representation, and pass to the limit. The result is that the log-price characteristic function keeps the exponential-affine form of Heston, with the Riccati equation replaced by a fractional Riccati equation of order $\alpha \in (1/2, 1)$, which has no closed form but is solved quickly by a fractional Adams scheme. A numerical illustration shows that the at-the-money implied volatility skew explodes as maturity goes to zero in the rough case and stays flat in the classical case. The contribution is a tractable pricing and calibration route for rough volatility, together with a microstructural origin for the model.

1 Introduction

1.1 Access note and method of review

CRITIQUE The review is based on the arXiv version 1 text (September 2016). The full text was read through a conversion tool that returned the paper in extracted and partly summarised form; the displayed equations of the introduction, the numerical section and the appendix were obtained in LaTeX, whereas Sections 2 to 4 and the proofs of Section 6 were obtained as structured descriptions. Where an equation could not be confirmed in its exact printed form it is marked as reconstructed. The paper PDF could not be downloaded to the library folder because the session network policy blocked file downloads, so no copy of the paper is stored next to this review. Numerical results labelled generated were produced by the reviewer and are not taken from the paper.

1.2 Source discipline used in this review

Passages tagged **PAPER** report what the authors state or prove. **EVIDENCE** reports what the mathematics or numerics establish. **CRITIQUE** is independent evaluation, **INFERENCE** follows from the paper without being stated, and **EXTENSION** goes beyond it.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks whether the log-price characteristic function $\mathbb{E}[e^{iaX_t}]$, with $X_t = \log(S_t/S_0)$, of a Heston-type model whose variance is driven by the kernel $(t-s)^{\alpha-1}$ with $\alpha \in (1/2, 1)$ can be written in a form that is as tractable as the classical Heston formula, so that Fourier option pricing and fast calibration remain available.

2.2 Motivation

Gatheral, Jaisson and Rosenbaum (2018) show that log-volatility behaves like a fractional Brownian motion with Hurst index H of order 0.1, and Bayer, Friz and Gatheral (2016) show that the same roughness fits the implied volatility surface, in particular the power-law explosion of the at-the-money skew at short maturities. The practical obstacle is computational: a non-Markovian, non-semimartingale volatility cannot be priced by PDE methods, and Monte Carlo is slow and delicate for calibration. A Heston-like characteristic function would remove that obstacle.

2.3 Existing literature and research gap

PAPER Heston (1993) obtains the characteristic function from the Feynman-Kac equation, using the Markov and time-homogeneous structure. That route is unavailable when $\alpha < 1$. The gap is therefore methodological: no pricing transform existed for rough stochastic volatility models, and the microstructural foundations of roughness (Jaisson and Rosenbaum 2016; El Euch, Fukasawa and Rosenbaum 2018) had not been exploited for pricing.

2.4 Hypothesis and claimed contribution

PAPER The central claim is Theorem 4.1: the characteristic function has the Heston structure with a fractional Riccati equation. **CRITIQUE** The claim is demonstrated as a theorem, so it is a mathematical result rather than a statistical hypothesis. What is only illustrated, and not demonstrated, is the claim that the resulting pricing method is efficient enough for practical calibration.

3 Theoretical Framework and Mechanism

3.1 The classical benchmark

PAPER The Heston model is

$$dS_t = S_t \sqrt{V_t} dW_t, \quad dV_t = \lambda(\theta - V_t) dt + \lambda\nu \sqrt{V_t} dB_t, \quad \langle dW, dB \rangle_t = \rho dt. \quad (1)$$

V_t Instantaneous variance of the log-price.

λ Speed of mean reversion of variance.

θ Long-run level of variance.

ν Volatility-of-volatility parameter; the diffusion coefficient of V is $\lambda\nu\sqrt{V}$.

ρ Correlation between price and variance shocks; it generates the leverage effect and the skew.

Applying Ito's formula to $L(t, a, V_t, S_t) = \mathbb{E}[e^{ia \log S_T} \mid \mathcal{F}_t]$ gives a Feynman-Kac equation, and the characteristic function takes the affine form $\mathbb{E}[e^{ia X_t}] = \exp[g(a, t) + V_0 b(a, t)]$ with

$$\square_t b = \frac{1}{2}(-a^2 - ia) + \lambda(ia\rho\nu - 1)b + \frac{(\lambda\nu)^2}{2}b^2, \quad b(a, 0) = 0, \quad g(a, t) = \theta\lambda \int_0^t b(a, s) ds. \quad (2)$$

Equation (2) is a Riccati equation: the term $\frac{1}{2}(-a^2 - ia)$ is the characteristic exponent of a Brownian motion with unit variance and the martingale drift $-\frac{1}{2}$, the linear term carries mean reversion and leverage, and the quadratic term comes from the square-root diffusion. It is solved in closed form, which is the origin of the Heston formula.

3.2 The rough Heston model

PAPER The authors replace the time integrals by convolutions with a power kernel:

$$dS_t = S_t \sqrt{V_t} dW_t, \quad V_t = V_0 + \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} \lambda (\theta - V_s) ds + \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} \lambda \nu \sqrt{V_s} dB_s. \quad (3)$$

For $\alpha = 1$ the kernel is constant and (3) is the integrated form of (1). The kernel is the one that appears in the Mandelbrot-van Ness representation of fractional Brownian motion,

$$W_t^H = \frac{1}{\Gamma(H+1/2)} \int_{-\infty}^0 [(t-s)^{H-1/2} - (-s)^{H-1/2}] dW_s + \frac{1}{\Gamma(H+1/2)} \int_0^t (t-s)^{H-1/2} dW_s, \quad (4)$$

so that $\alpha = H + 1/2$. The stochastic convolution has Holder regularity $\alpha - 1/2 - \varepsilon$ for any $\varepsilon > 0$, which the paper proves for the variance paths of (3). With $\alpha = 0.6$ the regularity is $0.1 - \varepsilon$, the empirical order of magnitude of Gatheral, Jaisson and Rosenbaum (2018). **CRITIQUE** The model is neither Markovian nor a semimartingale for $\alpha < 1$, so the state of the system at time t is the whole past of the driving noise, not (S_t, V_t) .

3.3 Microstructural construction: nearly unstable Hawkes processes

PAPER Since the Feynman-Kac route fails, the authors approximate (3) by point processes. Let $N^T = (N^{T,+}, N^{T,-})$ count upward and downward price ticks with intensity

$$\lambda_t^T = \hat{\mu}^T(t) \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \int_0^t \phi^T(t-s) \cdot d\mathbf{N}_s^T, \quad \phi^T = \phi^T \chi, \quad \chi = \frac{1}{\beta+1} \begin{pmatrix} 1 & \beta \\ 1 & \beta \end{pmatrix}. \quad (5)$$

$\hat{\mu}^T(t)$ Exogenous baseline arrival intensity of ticks.

ϕ^T Scalar kernel measuring how strongly a past tick excites future ticks.

β Asymmetry parameter in χ that controls the sign structure of feedback (leverage) between up and down ticks.

T Observation horizon that tends to infinity; time is rescaled by T .

Assumption 2.1 imposes $\phi^T = a^T \phi$ with $\phi = f_{\alpha,1}$ the Mittag-Leffler density and

$$a^T = 1 - \lambda T^{-\alpha}, \quad f_{\alpha,\lambda}(t) = \lambda t^{\alpha-1} E_{\alpha,\alpha}(-\lambda t^\alpha), \quad E_{\alpha,\beta}(z) = \sum_{n \geq 0} \frac{z^n}{\Gamma(\alpha n + \beta)}. \quad (6)$$

Two properties explain the choice. First, the spectral radius of $\int \chi$ equals one and $a^T \rightarrow 1$, so the process is *nearly unstable*: the branching ratio is close to one, meaning that almost every tick is endogenously triggered by an earlier one, which is the high-frequency counterpart of efficient markets and of the high share of order flow generated by algorithms. Second, the Mittag-Leffler

density has the tails

$$f_{\alpha,\lambda}(t) \sim \frac{\lambda}{\Gamma(\alpha)} t^{\alpha-1} \quad (t \rightarrow 0^+), \quad f_{\alpha,\lambda}(t) \sim \frac{\alpha}{\lambda \Gamma(1-\alpha)} t^{-(\alpha+1)} \quad (t \rightarrow \infty), \quad (7)$$

a power-law memory with tail index $\alpha < 1$ and an integrable singularity at the origin. Heavy tails in the kernel are what turn into the singular convolution kernel $(t-s)^{\alpha-1}$ of (3).

To obtain a nonzero initial variance in the limit, Assumption 2.2 uses the time-varying baseline

$$\hat{\mu}^T(t) = \mu T^{\alpha-1} + \xi \mu T^{\alpha-1} \left[\frac{1}{1-a^T} \left(1 - \int_0^t \phi^T(s) ds \right) - \int_0^t \phi^T(s) ds \right], \quad (8)$$

where $\mu > 0$ scales exogenous arrivals and $\xi > 0$ sets the initial volatility level. The rescaled objects, for $t \in [0, 1]$, are

$$X_t^T = \frac{1-a^T}{T^\alpha \mu} N_{tT}^T, \quad \Lambda_t^T = \frac{1-a^T}{T^\alpha \mu} \int_0^{tT} \lambda_s^T ds, \quad Z_t^T = \sqrt{\frac{T^\alpha \mu}{1-a^T}} (X_t^T - \Lambda_t^T), \quad (9)$$

the normalised counts, the normalised compensator and the rescaled martingale noise. **PAPER** Theorem 2.1 states that (Λ^T, X^T, Z^T) converges in law in the Skorokhod topology to (Λ, X, Z) with $X_t = \Lambda_t = \int_0^t Y_s ds (1, 1)^\top$, $Z_t = \int_0^t \sqrt{Y_s} dB_s$, and

$$Y_t = \xi + \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} \lambda(1-Y_s) ds + \sqrt{\frac{1+\beta^2}{\lambda \mu (1+\beta)^2}} \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} \sqrt{Y_s} dB_s, \quad (10)$$

where $B = (B^1 + \beta B^2) / \sqrt{1+\beta^2}$. Corollary 2.1 defines a microscopic log-price from the signed tick counts and shows it converges to $P_t = \int_0^t \sqrt{V_s} dW_s - \frac{1}{2} \int_0^t V_s ds$ with $V = \theta Y$, which is (3). **EVIDENCE** The proof proceeds by C -tightness (bounds on moments of the counts, Doob's inequality, vanishing jump size) and identification of the limit through a decomposition of Λ^T into a baseline term, a Mittag-Leffler convolution term and a martingale term; this review read the proof in summarised form only.

3.4 Characteristic function of a Hawkes process

PAPER For a d -dimensional Hawkes process with intensity $\lambda_t = \mu(t) + \int_0^t \phi(t-s) d\mathbf{N}_s$ and spectral radius of $\int \phi$ below one, Theorem 3.1 gives, through a cluster (branching) representation in which baseline arrivals are migrants and later events are their descendants,

$$\mathbb{E}[e^{i\mathbf{a} \cdot \mathbf{N}_t}] = \exp \left(\int_0^t (\mathbf{C}(\mathbf{a}, t-s) - \mathbf{1}) \cdot \mu(s) ds \right). \quad (11)$$

INFERENCE The vector $\mathbf{C}$ is the characteristic function of the size of the cluster generated by one migrant. By the independence of the sub-clusters of its direct offspring it must solve a fixed-point equation of the form $C_i(\mathbf{a}, t) = \exp(i a_i + \sum_j \int_0^t \phi_{ji}(s) (C_j(\mathbf{a}, t-s) - 1) ds)$. The text obtained for this review rendered the equation without the outer exponential, so the exponential form is a reconstruction by the reviewer from the branching property and should be checked against the paper. The

Wiener-Hopf lemma (Lemma A.1) then inverts linear Volterra equations $f = g + \varphi * f$ as $f = g + \psi * g$ with the resolvent $\psi = \sum_{k \geq 1} \varphi^{*k}$.

3.5 Main theorem: the fractional Riccati equation

PAPER Theorem 4.1 states that, for $\alpha \in (1/2, 1)$ and with the correlation restricted as the authors specify in the version read ($\rho \in (-1/\sqrt{2}, 1/\sqrt{2}]$), the characteristic function of the rough Heston log-price is

$$\mathbb{E}[e^{iaX_t}] = \exp\left[\theta\lambda I^1 b(a, t) + V_0 I^{1-\alpha} b(a, t)\right], \quad (12)$$

where $b(a, \cdot)$ solves the fractional Riccati equation

$$D^\alpha b(a, t) = \frac{1}{2}(-a^2 - ia) + \lambda(ia\rho\nu - 1)b(a, t) + \frac{(\lambda\nu)^2}{2}b^2(a, t), \quad I^{1-\alpha}b(a, 0) = 0. \quad (13)$$

The fractional integral and the Riemann-Liouville fractional derivative are

$$I^r f(t) = \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s) ds, \quad D^r f(t) = \frac{1}{\Gamma(1-r)} \frac{d}{dt} \int_0^t (t-s)^{-r} f(s) ds. \quad (14)$$

Equivalently, b solves the Volterra equation $b(a, t) = I^\alpha F(a, b(a, \cdot))(t)$, that is

$$b(a, t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} F(a, b(a, s)) ds, \quad F(a, x) = \frac{1}{2}(-a^2 - ia) + \lambda(ia\rho\nu - 1)x + \frac{(\lambda\nu)^2}{2}x^2. \quad (15)$$

Comparison with (2) shows three changes: the time derivative becomes D^α , the initial value V_0 multiplies $I^{1-\alpha}b$ instead of b , and the variance level enters through $I^1 b$ as before. When $\alpha = 1$, $D^1 = \square_t$, $I^0 = \text{id}$, and the Heston formula is recovered exactly.

3.6 Derivation: how the Hawkes limit produces the fractional Riccati equation

INFERENCE The paper's proof is long. The mechanism, reconstructed from the summary of Section 6, is as follows. Evaluate (11) at the rescaled argument $a_T \propto a T^{-\alpha/2} \sqrt{\cdot}$, so that $\mathbf{C} - \mathbf{1}$ is of order $T^{-\alpha}$. Writing $\mathbf{C} = \mathbf{1} + \mathbf{c}$, expanding the exponential to second order gives, for the scalar projection,

$$c + \frac{1}{2}c^2 + \dots = ia_T + \phi^T * c, \quad \text{so} \quad c = (\delta + \psi^T) * (ia_T + \frac{1}{2}c^2 + \dots), \quad \psi^T = \sum_{k \geq 1} (\phi^T)^{*k}. \quad (16)$$

The key step is the identity that, for $a^T = 1 - \lambda T^{-\alpha}$, the resolvent rescaled by T satisfies $\sum_{k \geq 1} (T\phi^T(T \cdot))^{*k} \rightarrow f_{\alpha, \lambda}$, the Mittag-Leffler density, whose Laplace transform is $\hat{f}_{\alpha, \lambda}(z) = \lambda/(\lambda + z^\alpha)$. The equation $b = f_{\alpha, \lambda} * F$ obtained in the limit is, after taking Laplace transforms, $\hat{b} = \lambda \hat{F}/(\lambda + z^\alpha)$, which is the Volterra form of a fractional differential equation of order α (Lemma A.2). Uniqueness of the continuous solution is shown by a Gronwall-type inequality for the Mittag-Leffler kernel (Lemma A.3), and the exponent α in the scaling $T^\alpha(\mathbf{Y}^T - \mathbf{1})$ is exactly what balances the surviving terms of order $T^{-\alpha}$ against negligible higher-order ones.

3.7 Consistency checks

INFERENCE Two direct checks of (12) and (13) are available without the proofs. For small t the solution behaves as $b(a, t) \approx F(a, 0) t^\alpha / \Gamma(\alpha + 1)$, so

$$\log \mathbb{E}[e^{iaX_t}] \approx V_0 t^{1-\alpha} \left[\frac{F(a, 0) s^\alpha}{\Gamma(\alpha + 1)} \right] (t) = V_0 F(a, 0) t = -\frac{V_0}{2}(a^2 + ia) t, \quad (17)$$

which is the characteristic function of a Brownian motion with variance rate V_0 and martingale drift, as required for any model with initial variance V_0 . Second, at $a = -i$ one has $-a^2 - ia = 0$, so $F(-i, 0) = 0$ and $b \equiv 0$ solves (13) with its initial condition; then $\mathbb{E}[e^{X_t}] = \mathbb{E}[S_t/S_0] = 1$, the martingale property, holds.

4 Data and Empirical Strategy

PAPER The paper contains no dataset. Its empirical content is a single numerical illustration (Section 5) of the pricing algorithm. The relevant design questions are therefore numerical, and the identifying variation is a controlled comparison between $\alpha = 1$ and $\alpha = 0.6$ with all other parameters held equal.

4.1 Numerical scheme

PAPER The Volterra equation (15) is solved by the fractional Adams predictor-corrector method of Diethelm, Ford and Freed (2002). On the grid $t_k = k\Delta$, the predictor and corrector are

$$\hat{b}^P(a, t_{k+1}) = \sum_{j=0}^k b_{j,k+1} F(a, \hat{b}(a, t_j)), \quad (18)$$

$$\hat{b}(a, t_{k+1}) = \sum_{j=0}^k a_{j,k+1} F(a, \hat{b}(a, t_j)) + a_{k+1,k+1} F(a, \hat{b}^P(a, t_{k+1})), \quad (19)$$

with weights $b_{j,k+1} = \frac{\Delta^\alpha}{\Gamma(\alpha+1)} [(k+1-j)^\alpha - (k-j)^\alpha]$, $a_{k+1,k+1} = \frac{\Delta^\alpha}{\Gamma(\alpha+2)}$, $a_{0,k+1} = \frac{\Delta^\alpha}{\Gamma(\alpha+2)} [k^{\alpha+1} - (k-\alpha)(k+1)^\alpha]$ and, for $1 \leq j \leq k$, $a_{j,k+1} = \frac{\Delta^\alpha}{\Gamma(\alpha+2)} [(k-j+2)^{\alpha+1} + (k-j)^{\alpha+1} - 2(k-j+1)^{\alpha+1}]$. The paper reports a global error of order $o(\Delta)$ and $o(\Delta^{2-\alpha})$ at interior points, and prices European options with the Lewis (2000) Fourier formula.

Table 1. Parameters of the numerical illustration as read from Section 5

Parameter	Value	Role
λ	2	Mean-reversion speed
ρ	-0.5	Price-variance correlation
V_0	0.4	Initial variance
ν	0.05	Volatility-of-volatility parameter
θ	0.04	Long-run variance level
α	1 and 0.6	Classical Heston and rough Heston ($H = 0.1$)

Source: El Euch and Rosenbaum (2016), Section 5, as extracted for this review; values should be checked against the printed table because the extraction tool did not return the original table layout.

4.2 Data limitations

CRITIQUE Because there is no market data, no calibration to observed surfaces, no timing comparison and no error table in the version read, claims about practical calibration speed are untested in the paper. The parameter values are stylised: with $V_0 = 0.4$ and $\theta = 0.04$ the initial volatility is about 63 percent and the long-run level is 20 percent, which is far from a typical equity index state, and $\nu = 0.05$ implies a small effective vol-of-vol $\lambda\nu = 0.1$.

5 Results

5.1 Main theoretical result

EVIDENCE Theorem 4.1 establishes (12) and (13) under the stated assumptions. Its economic meaning is that the model remains affine in the sense of Fourier pricing: one needs a single numerical solution of a one-dimensional fractional ODE per Fourier frequency a , the same cost structure as Heston with a Riccati solve, but a heavier solve because the fractional equation has memory and a quadratic number of operations in the number of time steps.

5.2 Reported numerical result

PAPER With the parameters of Table 1, the at-the-money skew, defined as the derivative of implied volatility with respect to log-strike, explodes as maturity tends to zero in the rough case and remains flat in the classical case. **EVIDENCE** This is the behaviour that the theory of rough volatility predicts: the short-maturity skew scales as $T^{H-1/2} = T^{\alpha-1}$ (Alos, Leon and Vives 2007; Fukasawa 2011; Bayer, Friz and Gatheral 2016), which equals $T^{-0.4}$ for $\alpha = 0.6$ and T^0 for $\alpha = 1$. The paper states the finding in words only in the version read; no numerical table of skews accompanies it.

5.3 Independent replication (generated)

EXTENSION To check the result, the reviewer implemented the fractional Adams scheme above, the formula (12) with a product-integration rule for $I^{1-\alpha}h$, and Lewis (2000) inversion with Black-Scholes implied volatility, using the parameters of Table 1 and taking the numbers as read, not as confirmed. The results are generated by the reviewer and are not official or published values. The skew is reported in absolute value with respect to log-strike $\log(K/S)$; its sign is negative because $\rho < 0$.

Table 2. Generated replication: at-the-money implied volatility and absolute skew by maturity

Maturity (years)	ATM vol, $\alpha = 0.6$	ATM vol, $\alpha = 1$	Skew , $\alpha = 0.6$	Skew , $\alpha = 1$
0.02	0.596	0.627	0.128	0.020
0.05	0.574	0.618	0.086	0.020
0.10	0.549	0.605	0.063	0.020
0.20	0.517	0.580	0.045	0.020
0.40	0.479	0.535	0.032	0.020
0.70	0.444	0.482	0.023	0.020
1.00	0.421	0.440	0.019	0.019

Source: generated by the reviewer with the fractional Adams scheme (80 to 240 steps) and Lewis (2000) inversion; parameters from Table 1; skew by central finite difference in log-strike with step 0.02. Not an official or published result.

Figure 1. At-the-money skew as a function of maturity, rough and classical Heston

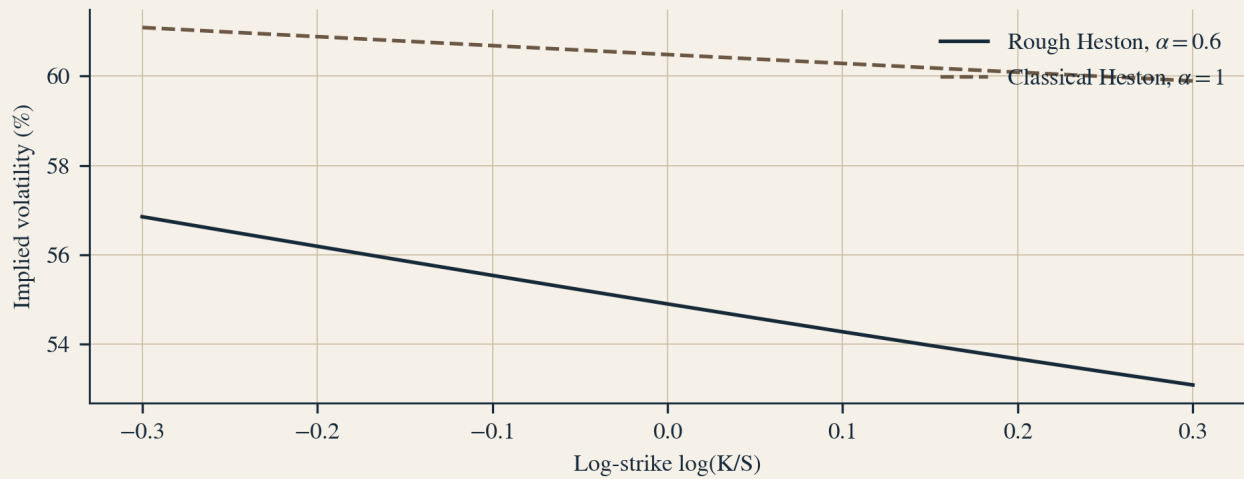


Source: generated by the reviewer; log scales; maturities 0.02 to 1 year; parameters of Table 1; dotted line is the theoretical short-maturity slope $T^{\alpha-1} = T^{-0.4}$ anchored at $T = 0.1$.

EVIDENCE Table 2 answers the question whether the Fourier machinery of Theorem 4.1 reproduces the rough skew. The dependent quantity is the implied-volatility slope, the key column is the absolute skew for $\alpha = 0.6$. At $T = 0.02$ the rough skew is 0.128 against 0.020 for Heston, a factor of 6.4. A regression of log skew on log maturity over $T \leq 0.4$ gives a slope of -0.46 for $\alpha = 0.6$, against the theoretical -0.40 , and $+0.003$ for $\alpha = 1$. The difference from -0.40 is plausibly due to finite-maturity corrections and to the coarse grid at the shortest maturities (an inference, not tested here). The table

does not establish accuracy against an independent reference such as Monte Carlo, so the replication confirms the qualitative structure, not the numerical error bounds.

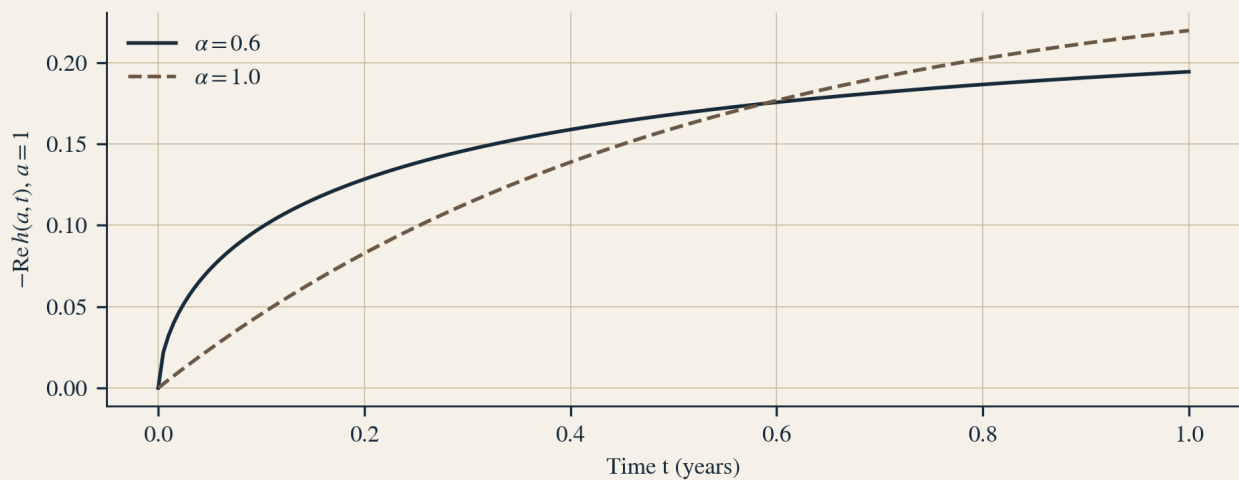
Figure 2. Implied volatility smile at three-week to one-month maturity, rough and classical Heston



Source: generated by the reviewer; maturity 0.1 year; parameters of Table 1; log-strike on the horizontal axis.

The smile in the figure above shows the leverage-induced asymmetry in both models, steeper for $\alpha = 0.6$. It is a smile at one maturity and cannot by itself distinguish roughness from other sources of skew; the discriminating information is the term structure of the skew in the previous figure.

Figure 3. Solution of the Riccati and fractional Riccati equations at $a = 1$



Source: generated by the reviewer; $t \in [0, 1]$; negative of the real part of $h(1, t)$; parameters of Table 1.

The fractional solution rises as t^α at the origin, faster than the linear Heston rise, then flattens. This short-time behaviour is the origin of the heavier short-maturity skew: the exponent in $h \sim t^\alpha$ feeds directly into the exponent of the skew.

6 Robustness and Additional Tests

Table 3. Robustness: concern, test in the paper, result, and whether it resolves the concern

Concern	Test	Result	Resolved?
Is the limit model well defined and rough?	Tightness and Holder regularity proof, Theorem 2.1	Paths have regularity $\alpha - 1/2 - \varepsilon$	Yes, as a theorem
Is the fractional Riccati equation well posed?	Gronwall-type argument, Lemma A.3	Unique continuous solution	Yes, as a theorem
Does the formula nest Heston?	Set $\alpha = 1$	Classical Riccati recovered	Yes
Is the numerical scheme convergent?	Error analysis of the Adams method	$o(\Delta)$ global, $o(\Delta^{2-\alpha})$ interior	Partly: rates cited, not verified on this equation
Does the correlation restriction bind?	None reported	$\rho \in (-1/\sqrt{2}, 1/\sqrt{2}]$ in the version read	No
Is the method fast enough for calibration?	Qualitative remark only	No timings in the version read	No
Does the model fit market data?	None in this paper	Relies on earlier work	No

Source: reviewer’s assessment of El Euch and Rosenbaum (2016).

CRITIQUE The robustness that matters for a theorem is mathematical, and the paper handles it by proofs. What is missing is numerical robustness: sensitivity of the solution to the grid, to the truncation of the Fourier integral, and to complex arguments with large $|a|$, where the quadratic nonlinearity can cause the fractional Adams iteration to lose accuracy.

7 Critical Evaluation

ESTABLISHED

Under the paper’s assumptions, the rough Heston log-price has a characteristic function of the Heston exponential-affine form with a fractional Riccati equation, and the model is a scaling limit of nearly unstable Hawkes processes. The classical Heston result is the special case $\alpha = 1$.

SUPPORTED BUT CONDITIONAL

Fast Fourier pricing is feasible: the independent replication solves the equation and inverts prices in well under a second per maturity on a standard computer, but with a stylised parameter set and without an external accuracy benchmark.

SUGGESTIVE

That the microstructural construction explains why markets are rough. The Hawkes system is one model consistent with the stylised facts, so the link is an existence argument, not an identification of the actual mechanism.

NOT ESTABLISHED

That the model fits market option surfaces, that the correlation restriction is innocuous, and that the parameters implied by the Hawkes microstructure $(\lambda, \mu, \beta, \nu)$ correspond to observed order-flow quantities.

7.1 Strengths

RECOGNISED BY THE AUTHORS

The result restores the Heston-type workflow (characteristic function, Fourier inversion, fast calibration) for rough models, and the proof technique yields microstructural foundations at the same time.

IDENTIFIED INDEPENDENTLY

The paper turns a modelling obstacle into a computational problem of known type: solving a one-dimensional Volterra equation, for which stable schemes exist. Its approach is also reusable: the limit-of-Hawkes argument was later extended by the same group (El Euch and Rosenbaum 2018 on hedging) and by affine Volterra theory (Abi Jaber, Larsson and Pulido 2019). The model also nests Heston, which makes the comparison clean.

7.2 Weaknesses

Single most important weakness: the gap between the theorem and calibrated practice

PROBLEM. No evidence of calibration quality or computational cost is provided, and the numerical parameter set is far from a market state.

WHY IT MATTERS. The practical value of the paper rests on speed and accuracy of the Fourier solve across strikes and maturities, in particular for the short maturities where roughness matters most.

POTENTIAL BIAS. Overstated practicality; the quadratic cost in the number of time steps and the accuracy loss near $t = 0$, where $h \sim t^\alpha$ is non-smooth, can degrade accuracy exactly where the skew is largest.

HOW IT COULD BE TESTED. Compare the Adams solution with a high-precision reference and with multifactor Markovian approximations across a grid of (a, T) , and calibrate to an observed SPX surface with timing.

Further weaknesses

CORRELATION RESTRICTION. The version read restricts ρ to $(-1/\sqrt{2}, 1/\sqrt{2}]$, which excludes the strongly negative correlations typical of equity indices; if this is not relaxed, the theorem does not cover the empirically relevant region. How to test: check whether the published version lifts it, and whether the numerical formula remains accurate for $\rho < -0.71$.

WEAK CONVERGENCE IS NOT A PRICING GUARANTEE. Convergence in law of the processes does not by itself give convergence of option prices or exact equality of characteristic functions for the limit model; the paper handles the characteristic function directly, but extensions to path-dependent payoffs need extra arguments.

NONNEGATIVE VARIANCE AND ARBITRAGE. The limit is of positive intensities, so positivity of V is inherited, but the discussion of existence and uniqueness in law of the SDE is not independent of the construction.

PARAMETER INTERPRETATION. The map from Hawkes parameters to $(\lambda, \nu, \theta, \rho)$ implies restrictions on ν and ρ through β and μ ; the paper does not discuss whether market-implied values respect them.

8 Contradictions and Edge Cases

CRITIQUE Four limits are informative. As $\alpha \rightarrow 1$ the kernel becomes constant and the formula tends to the Heston one, as verified above. As $\alpha \rightarrow 1/2$, $H \rightarrow 0$ and the regularity $\alpha - 1/2 - \varepsilon$ degenerates; the Volterra kernel $(t-s)^{-1/2}$ has an integrable but sharply peaked singularity and the formula for $\theta \lambda^1 b$ remains defined, but the approximating Hawkes system requires $\alpha > 1/2$ for the square-integrability used in the martingale part, so the theorem does not apply at the boundary. For large $|a|$ the quadratic term $\frac{(\lambda \nu)^2}{2} b^2$ dominates and complex solutions may blow up in finite time for some parameter choices; in the Heston case this is controlled by the explicit solution (and by the choice of branch discussed by Albrecher et al. 2007), whereas in the rough case no analogue is proved in the version read. For $\nu \rightarrow 0$ the quadratic term disappears and b is the solution of a linear fractional equation expressed through $E_{\alpha, \alpha}$, so the variance is deterministic and no skew arises from roughness alone: roughness without vol-of-vol cannot produce an explosive skew, which is why the skew level in the replication table is small even at $T = 0.02$ with $\nu = 0.05$. A different mechanism could generate the same short-maturity skew: Alos, Leon and Vives (2007) show $T^{-1/2}$ -type behaviour in jump-diffusion stochastic volatility models, and price jumps of finite activity produce a skew diverging as $T^{-1/2}$, so the power $T^{-0.4}$ is difficult to separate from $T^{-1/2}$ with noisy short-maturity quotes (inference).

9 Further Literature

The table below places the paper among its closest neighbours, ordered from the foundation to critiques and extensions.

Table 4. Literature connections

Paper	Shared idea	Relationship	Why it matters
Heston (1993)	Characteristic function of stochastic volatility	Preceded; recovered at $\alpha = 1$	Benchmark and template
Gatheral, Jaisson and Rosenbaum (2018)	Volatility is rough, $H \approx 0.1$	Empirical motivation	Supplies the target model
Bayer, Friz and Gatheral (2016)	Pricing under rough volatility by simulation	Alternative pricing route	Monte Carlo benchmark
Jaisson and Rosenbaum (2015, 2016)	Nearly unstable Hawkes processes and rough limits	Technical foundation	Source of the limit theorem
El Euch, Fukasawa and Rosenbaum (2018)	Microstructure of leverage and roughness	Companion paper	Economic foundation
Abi Jaber, Larsson and Pulido (2019)	Affine Volterra processes	Generalises the result	Direct proof without Hawkes approximation
Abi Jaber and El Euch (2019)	Multifactor Markovian approximation	Improves speed	Alternative numerics for calibration
Fukasawa, Takabatake and Westphal (2022)	Is volatility rough?	Questions the empirical premise	Possible contradiction
Bennedsen, Lunde and Pakkanen (2022)	Decoupling short and long memory	Challenges single power kernel	Alternative kernels

Source: reviewer's classification; relationship descriptions are the reviewer's own.

10 Research Gaps

10.1 Paper-level gaps

Gap. Calibration performance and numerical error control. *Why it matters:* it decides practical adoption. *Existing limitation:* one parameter set and no benchmark. *Way forward:* systematic error maps and timing against alternatives.

Gap. Correlation range and dependence on the Hawkes parametrisation. *Why it matters:* equity indices have strongly negative correlation. *Existing limitation:* stated restriction in the version read. *Way forward:* direct proofs via Volterra affine theory.

10.2 Literature-level gaps

Gap. Joint fit of the SPX and VIX surfaces. *Why it matters:* a single-factor rough Heston model struggles to match both. *Existing limitation:* the model has too few degrees of freedom for term-structure shape. *Way forward:* quadratic rough Heston and multifactor extensions.

10.3 Methodological gaps

Gap. Estimating roughness from options and from realised variance consistently. *Why it matters:* implied and historical H need not coincide. *Existing limitation:* noisy proxies for volatility. *Way*

forward: high-frequency data with microstructure noise correction, and joint likelihood on both measures.

11 Research Opportunities

Idea 1. Joint SPX and VIX calibration with fractional Riccati pricing

RESEARCH QUESTION. Can a rough Heston model with a forward-variance curve, priced with the fractional Riccati characteristic function, fit SPX and VIX smiles simultaneously better than classical two-factor models?

MOTIVATION. The joint calibration problem is the standard test of any volatility model.

HYPOTHESIS. Roughness improves the short-maturity fit of both surfaces at equal parameter count.

ECONOMIC MECHANISM. Endogenous order flow (near-unit branching ratio) makes variance a long-memory process.

LITERATURE CONNECTION. Gatheral, Jusselin and Rosenbaum (2020); Horvath, Jacquier and Tankov (2020).

DATA. OptionMetrics SPX options and CBOE VIX options, 2010 to 2025.

VARIABLES. Implied volatilities by strike and maturity, forward variance curve, VIX futures.

METHOD. Calibration by least squares on both surfaces with fractional Adams solves.

IDENTIFICATION STRATEGY. Out-of-sample surface prediction across days and stress periods.

EXPECTED CONTRIBUTION. A benchmark for cost and fit of rough versus classical models.

MAIN RISK. Overfitting and computational cost.

Idea 2. Is implied roughness equal to historical roughness?

RESEARCH QUESTION. Is the Hurst index implied by the short-maturity skew, H^{imp} , equal to the index estimated from high-frequency realised variance, H^{hist} , across assets and periods?

MOTIVATION. The equality is implied by the model but untested in this paper; its failure would indicate a roughness risk premium or model misspecification.

HYPOTHESIS. H^{imp} exceeds H^{hist} , because skew exponents are contaminated by jumps (Alos, Leon and Vives 2007).

ECONOMIC MECHANISM. Risk premia and jumps alter the pricing measure exponent without changing historical roughness.

LITERATURE CONNECTION. Fukasawa, Takabatake and Westphal (2022); Gatheral, Jaisson and Rosenbaum (2018).

DATA. Five-minute realised variance for S&P 500 constituents and index options, 2012 to 2025.

VARIABLES. Log-realised variance increments, short-maturity ATM skew, jump indicators.

METHOD. Scaling of variogram moments for H^{hist} ; regression of log skew on log maturity for H^{imp} .

IDENTIFICATION STRATEGY. Cross-sectional variation across assets and time, with jump-robust skew estimators.

EXPECTED CONTRIBUTION. Direct test of the model's two-measure consistency.

MAIN RISK. Measurement error in short-maturity quotes.

Idea 3. Rough Heston for cryptocurrency options

RESEARCH QUESTION. Do bitcoin and ether option surfaces display a more extreme short-maturity skew exponent than equity options?

MOTIVATION. Trading is continuous, retail-driven and highly self-exciting, so the Hawkes branching ratio may be nearer one.

HYPOTHESIS. Estimated α is smaller than for equities.

ECONOMIC MECHANISM. Higher endogeneity of order flow.

LITERATURE CONNECTION. Jaisson and Rosenbaum (2016); Hardiman, Bercot and Bouchaud (2013).

DATA. Deribit option quotes and tick trades, 2020 to 2025.

VARIABLES. Implied vol surface, branching ratio estimated from tick data.

METHOD. Calibrate rough Heston by fractional Riccati; estimate Hawkes kernels by maximum likelihood.

IDENTIFICATION STRATEGY. Compare across assets and liquidity regimes.

EXPECTED CONTRIBUTION. A new asset class for the roughness literature.

MAIN RISK. Market fragmentation and thin short-dated quotes.

Idea 4. Numerical error control for fractional Riccati pricing

RESEARCH QUESTION. What is the pricing error of the fractional Adams scheme as a function of (a, T, α, ν) against a high-precision reference?

MOTIVATION. The paper cites convergence rates but does not provide error maps.

HYPOTHESIS. Errors concentrate at short maturities and large $|a|$ and are reduced by graded meshes.

ECONOMIC MECHANISM. Not applicable; a computational question.

LITERATURE CONNECTION. Diethelm, Ford and Freed (2002); Abi Jaber and El Euch (2019).

DATA. Simulated parameter grids.

VARIABLES. Absolute price error in implied volatility basis points, runtime.

METHOD. Convergence studies against Monte Carlo and multifactor schemes.

IDENTIFICATION STRATEGY. Controlled grid refinement.

EXPECTED CONTRIBUTION. Practical guidance for practitioners.

MAIN RISK. Limited novelty.

Idea 5. Roughness and the branching ratio over time

RESEARCH QUESTION. Does the estimated near-instability of order flow predict changes in the roughness exponent of volatility across market regimes?

MOTIVATION. In the paper the exponent is tied to the tail of the Hawkes kernel and the distance of the branching ratio from one.

HYPOTHESIS. Higher endogeneity ratios coincide with lower α in realised-variance data.

ECONOMIC MECHANISM. Reflexive order flow lengthens memory.

LITERATURE CONNECTION. Hardiman, Bercot and Bouchaud (2013); El Euch, Fukasawa and Rosenbaum (2018).

DATA. Level-one tick data for liquid futures, 2015 to 2025.

VARIABLES. Branching-ratio estimates, realised-variance Hurst estimates by month.

METHOD. Rolling Hawkes maximum likelihood and variogram estimation, then panel regression.

IDENTIFICATION STRATEGY. Exogenous liquidity shocks, such as tick-size changes, as instruments.

EXPECTED CONTRIBUTION. Empirical test of the microstructural foundation.

MAIN RISK. Estimation noise in both quantities.

11.1 Ranking

The five ideas are scored below on novelty, feasibility, data availability, methodological quality, importance and potential contribution.

Table 5. Ranking of research ideas (reviewer scores from 1 to 5; higher is better)

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Joint SPX and VIX	3	3	4	4	5	4	23
2. Implied versus historical roughness	4	4	4	4	5	4	25
3. Crypto options	4	4	3	3	3	4	21
4. Numerical error control	3	5	5	4	3	2	22
5. Roughness and branching ratio	5	3	3	4	4	5	24

Source: reviewer's own scoring; subjective and unweighted.

The top three are Idea 2, Idea 5 and Idea 1. Idea 2 is cheap and directly tests a joint implication of the model; Idea 5 is the most novel because it tests the microstructural foundation that is the paper's distinctive claim; Idea 1 has the highest practical importance.

12 Learning Framework

12.1 Prerequisite concepts

Ito calculus and Feynman-Kac; the Heston model and its Riccati equation; Fourier option pricing (Lewis, Carr-Madan); fractional Brownian motion and the Hurst index; Laplace transforms and convolutions; point processes and the Hawkes process.

12.2 Core concepts learned

- Rough volatility enters through a power kernel, and its roughness H equals $\alpha - 1/2$.
- Nearly unstable Hawkes processes with heavy-tailed kernels converge to rough variance dynamics.
- The characteristic function stays affine, with a fractional Riccati equation replacing the Riccati equation.

12.3 Important equations

Equations (3), (10), (12), (13) and (15) are the ones to retain.

12.4 Methods learned

Cluster representation of Hawkes processes; scaling limits with a renormalised nearly critical kernel; resolvent and Wiener-Hopf inversion of Volterra equations; the fractional Adams predictor-corrector scheme; Lewis Fourier inversion of option prices.

12.5 Questions to test understanding

COMPREHENSION

1. Why does $\alpha = 0.6$ correspond to a Hurst index of 0.1, and what does that imply for path regularity?
2. What changes between the Heston Riccati equation and the fractional Riccati equation?

ANALYSIS

1. Why does a branching ratio close to one with heavy-tailed kernel give a rough, not a smooth, limit?
2. Show that $b \equiv 0$ solves (13) at $a = -i$ and explain the martingale interpretation.

RESEARCH

1. How would you test whether the skew exponent in data is -0.4 rather than -0.5 ?
2. How would you extend the result to $\rho \leq -1/\sqrt{2}$?

12.6 Papers to read next

Gatheral, Jaisson and Rosenbaum (2018); Bayer, Friz and Gatheral (2016); Abi Jaber, Larsson and Pulido (2019); El Euch and Rosenbaum (2018) on perfect hedging; Gatheral, Jusselin and Rosenbaum (2020).

13 Conclusion

13.1 One-sentence thesis

The log-price characteristic function of the rough Heston model keeps Heston's exponential-affine form with a fractional Riccati equation, so Fourier pricing is available for rough volatility, and the model arises as the limit of nearly unstable Hawkes processes.

13.2 Three most important insights

- Roughness is a kernel property, and the kernel can be generated by near-critical self-exciting order flow.
- Tractability survives the loss of the Markov property because the Hawkes structure is branching.
- The short-maturity skew exponent $\alpha - 1$ links the roughness parameter directly to option prices.

13.3 Strongest evidence

The theorem itself, and its exact reduction to Heston at $\alpha = 1$; the generated replication reproduces the qualitative skew behaviour.

13.4 Biggest weakness

The absence of calibration, accuracy and timing evidence, together with the correlation restriction in the version read.

13.5 Most important research gap

Whether roughness estimated from options matches roughness estimated from historical volatility, and whether it is stable over regimes.

13.6 Best research opportunity

A joint test of implied and historical roughness on high-frequency equity data (Idea 2).

13.7 Three papers to read next

Gatheral, Jaisson and Rosenbaum (2018); Abi Jaber, Larsson and Pulido (2019); Fukasawa, Takabatake and Westphal (2022).

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